

BRC WRF First Case Runbook

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This is the current start-to-finish proof path that connects `../brc-tools` input staging to WPS, `real.exe`, `wrf.exe`, and archive checks for this `brc-wrf` checkout.

It is a runbook, not an approval to submit jobs. DTN staging jobs, WPS work, `real.exe`, `wrf.exe`, scaling sweeps, and Slurm submissions still need explicit human approval before they run.

Current Status

- Validated path: NAM-only, single WPS stream, `vtable.NAM`, `interval_seconds = 21600`.
- Validated case: known Jan-2013 Uinta Basin 12/4 km nested domain, `2013-01-31_12:00:00` through `2013-02-02_00:00:00`.
- Validated runtime target: `notch392`, `lawson-np`, one node, 56 tasks, Intel MPI launched with `srun --mpi=pmi2`.
- Not yet validated: GEFSv12 reforecast plus NAM two-stream forcing (`fg_name = 'GEFS', 'NAM'`, `interval_seconds = 10800`).

Do not describe this proof as a fresh standalone 4 km case or as a completed GEFS+NAM two-stream run.

The input contract/source identity for the proof is NAM. The old proof scratch `namelist.wps` used the WPS default intermediate prefix, so its live values are `ungrib prefix = 'FILE'` and `metgrid fg_name = 'FILE'`. Future NAM-only runs may use `prefix = 'NAM'` and `fg_name = 'NAM'` if those names are kept paired.

Ownership

STEP	OWNER	NODE CLASS	MAIN OUTPUT
Plan and stage GRIB input	<code>../brc-tools</code>	DTN	<code>/scratch/general/vast/\$USER/wrf_inputs/<case>/</code>
Validate staged input	<code>../brc-tools</code>	login or DTN	<code>manifest_<case>.json</code> check results
WPS setup and run	WPS plus this runbook	allocation or batch	<code>met_em.d0*</code>
<code>real.exe</code> and <code>wrf.exe</code>	<code>brc-wrf</code> build/run tree	<code>notch392</code> batch	<code>wrfinput_d0*</code> , <code>wrfbdy_d01</code> , <code>wrfout_d0*</code>
Archive and explain result	<code>brc-wrf</code> wrapper	end of batch	<code>lawson-group6/</code> <code><namespace>/</code> <code>wrf_archive/<case>/</code> <code>run_<UTC>/</code>

brc-tools Input Contract

Read `../brc-tools/docs/WRF-INPUT-STAGING.md` before consuming staged inputs. Fresh `brc-tools` staging writes both:

```
/scratch/general/vast/$USER/wrf_inputs/<case>/manifest_<case>.json
/scratch/general/vast/$USER/wrf_inputs/<case>/contract_<case>.json
```

Use the contract as the input handshake. It records the staged source counts, cadence, valid window, suggested WPS `fg_name` , and `interval_seconds` .

For the validated NAM-only path, the WPS-side constants are:

FACT	VALUE
<code>wps_fg_name</code>	<code>NAM</code>
observed proof <code>namelist.wps</code> <code>fg_name</code>	<code>FILE</code>
<code>interval_seconds</code>	<code>21600</code>
<code>num_metgrid_levels</code>	<code>40</code>
<code>met_em</code> count	14 files, d01 and d02 at 6-hour cadence
Required fields observed	<code>LANDSEA</code> , <code>SOILHGT</code> , <code>SKINTEMP</code> , <code>SEAICE</code> , <code>SNOW</code> , <code>SNOWH</code> , four soil-temperature layers, four soil-moisture layers

The old proof scratch set was staged before the contract sidecar existed, so it may have only the manifest. This `brc-wrf` checkout carries `brc-cases/jan2013_basin_nam.contract.json` , a reconstructed NAM-only contract for the validated consumption path, so the case validator has a

durable handshake to read. Do not treat the missing old scratch sidecar as evidence that fresh `brc-tools` staging lacks the contract feature.

Cheap checks that do not submit jobs:

```
cd ~/gits/brc-tools
python scripts/stage_wrf_inputs.py --plan --case jan2013_basin_gefs \
  --init-time "2013-01-31 12Z" --source nam_analysis
python scripts/stage_wrf_inputs.py --verify-manifest \
  /scratch/general/vast/$USER/wrf_inputs/jan2013_basin_gefs/manifest_jan2013_basin_gefs.json
```

Current checked evidence: the existing staged proof manifest verifies 28/28 OK for 7 NAM files plus 21 optional GEFS reforecast files. The validated WPS run consumed the NAM stream only.

WPS Handoff

Stage active case I/O under scratch:

```
/scratch/general/vast/$USER/wrf_runs/<case>/
  grib_data/
  wps_run/
  wrf_run/
```

For the validated path:

- Link or copy `wrf_inputs/<case>/nam_analysis/` into `grib_data/`.
- Use `vtable.NAM`.
- Keep the WPS intermediate prefix and metgrid `fg_name` paired. The checked proof scratch used `prefix = 'FILE'` and `fg_name = 'FILE'`; a cleaned-up NAM-named stream should use `prefix = 'NAM'` and `fg_name = 'NAM'`.
- Set WPS/WRF interval to 6 hours (`interval_seconds = 21600`).
- Keep `geog_data_path = /uufs/chpc.utah.edu/common/home/lawson-group6/WPS_GEOG/`.

After WPS, check that `met_em` exists for both domains and that the resulting WRF namelist uses `num_metgrid_levels = 40`.

WRF Run Wrapper

Use the CHPC-validated wrapper as the starting point:

```
../brc-knowledge/scholarium/reference-base/resources/run_wrf_feb05.slurm
```

The wrapper must keep these CHPC-specific details:

- reload the validated Intel/HDF5/netCDF module stack inside the batch job;
- run `real.exe` directly and require `SUCCESS COMPLETE REAL_EM INIT`;
- move `real.exe` `rs1.*` files aside before `wrf.exe`;
- launch WRF with `srun --mpi=pmi2 -n "$SLURM_NTASKS" ./wrf.exe`;
- require `SUCCESS COMPLETE WRF` in `rs1.out.0000`.

WRF output filenames contain colons, so archive commands must prefix local sources or use absolute paths:

```
rsync -av ./wrfout_d0* "$ARCHIVE_DIR/"
rsync -av namelist.input rsl.out.0000 rsl.error.0000 \
    real.rsl.out.0000 real.rsl.error.0000 "$ARCHIVE_DIR/"
```

Treat Slurm batch state, the WRF .0 step state, WRF success markers, and archive completeness as separate facts. A post-WRF archive failure can mark the batch failed even when WRF completed successfully.

Proof Evidence

Evidence checked on 2026-06-13 without submitting a new job:

- `brc-tools` manifest verification: `/scratch/general/vast/u0737349/wrf_inputs/jan2013_basin_gefs/manifest_jan2013_basin_gefs.json` verified 28/28 OK.
- Archive: `/uufs/chpc.utah.edu/common/home/lawson-group6/jrlawson/wrf_archive/jan2013_basin_gefs/run_20260613T044846Z` contains 194 files, including 74 `wrfout_d0*` files, and is 2.2G.
- `real.rsl.out.0000` contains `d01 2013-02-02_00:00:00 real_em: SUCCESS COMPLETE REAL_EM INIT.`
- `rsl.out.0000` contains `d01 2013-02-02_00:00:00 wrf: SUCCESS COMPLETE WRF.`

The non-fatal `real.exe` soil message observed for the proof was: `forcing artificial silty clay loam at 2 points, out of 18000.`

Next Tests

1. GEFS+NAM two-stream WPS/real path: build or select a GEFSv12 reforecast Vtable, ungrib GEFS and NAM separately, run metgrid with `fg_name = 'GEFS', 'NAM'`, then prove `real.exe`.
2. Scaling sweep on `notch392`: run the same case at 16, 28, and 56 tasks, then record wall time per simulated hour and peak memory.
3. Use `brc-cases/` to review the case manifest, validate cheap metadata, and render Slurm text before any submitted run.
4. Promote the run wrapper into a maintained `brc-wrf-side` template only after the exact build/WPS/run directory contract is settled.